

Steps that I had followed to Implement Data Virtualization

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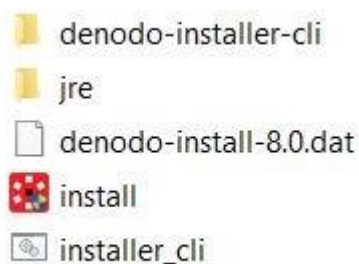
Installing Denodo:

About Denodo:

Denodo is a data virtualization tool wherein we can extract data from different data sources.

First download the [Denodo Express](#) installation package from your user account.

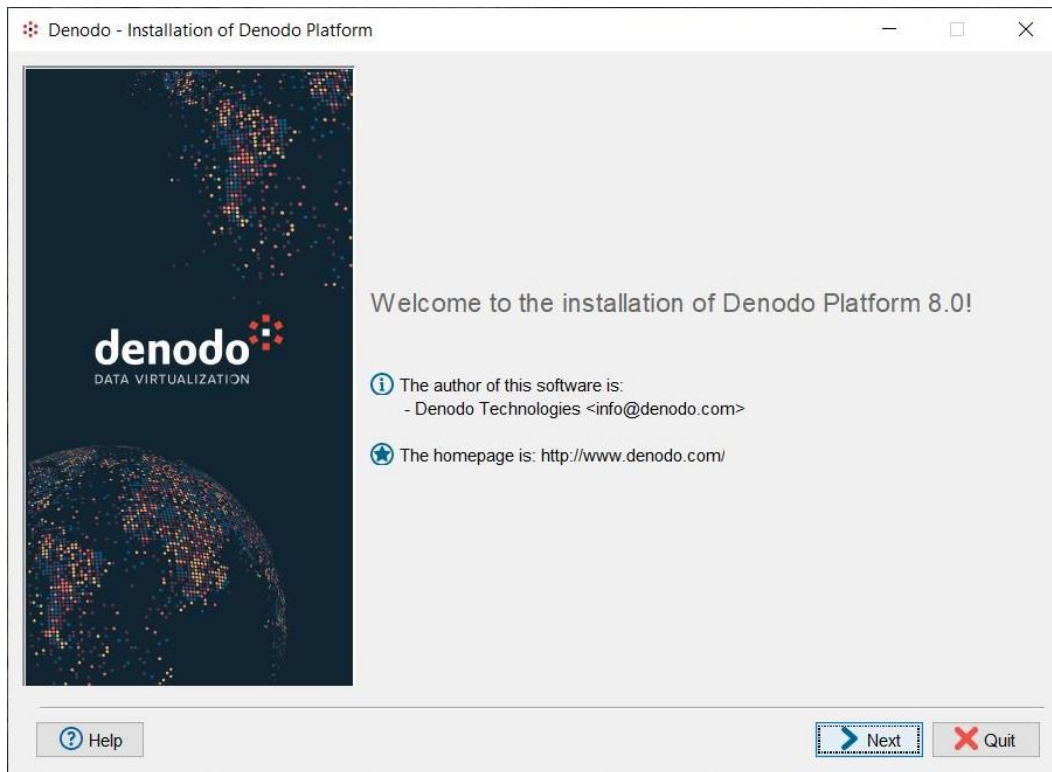
The installation package is a .zip file. After decompressing the package you will see the files shown in the image below:



To start the installation GUI:

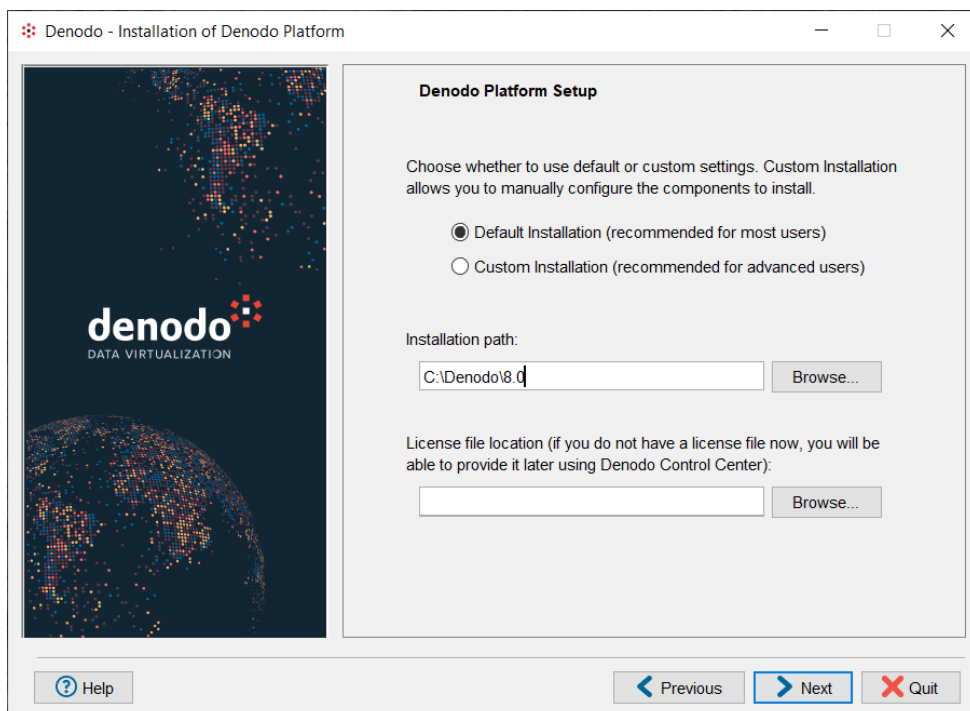
- **Windows OS:** right-click the *install.exe* file and select Run as administrator.
- **Linux OS:** execute *install.sh*.

The installer will show up.

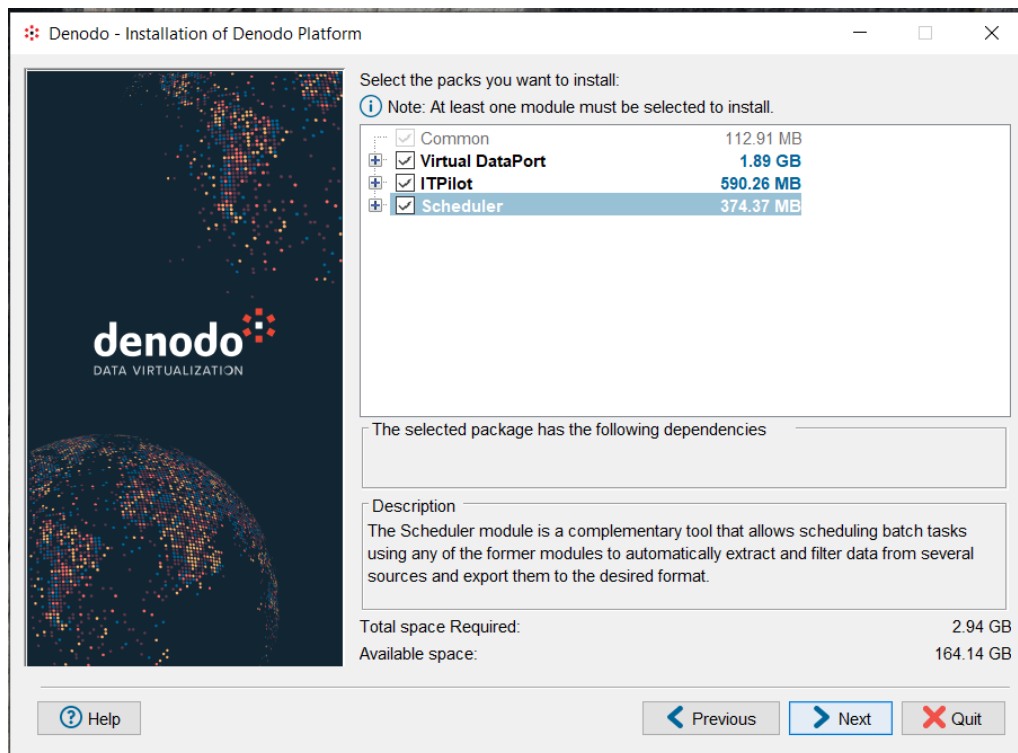


After accepting the terms of the license, you have to select the installation directory (for example: `C:\Denodo\8.0` or `/opt/denodo/8.0`).

If you already have a Denodo license file, you can select it clicking on the "Browse" button. Otherwise, you can install the license later from the **Denodo Control Center**



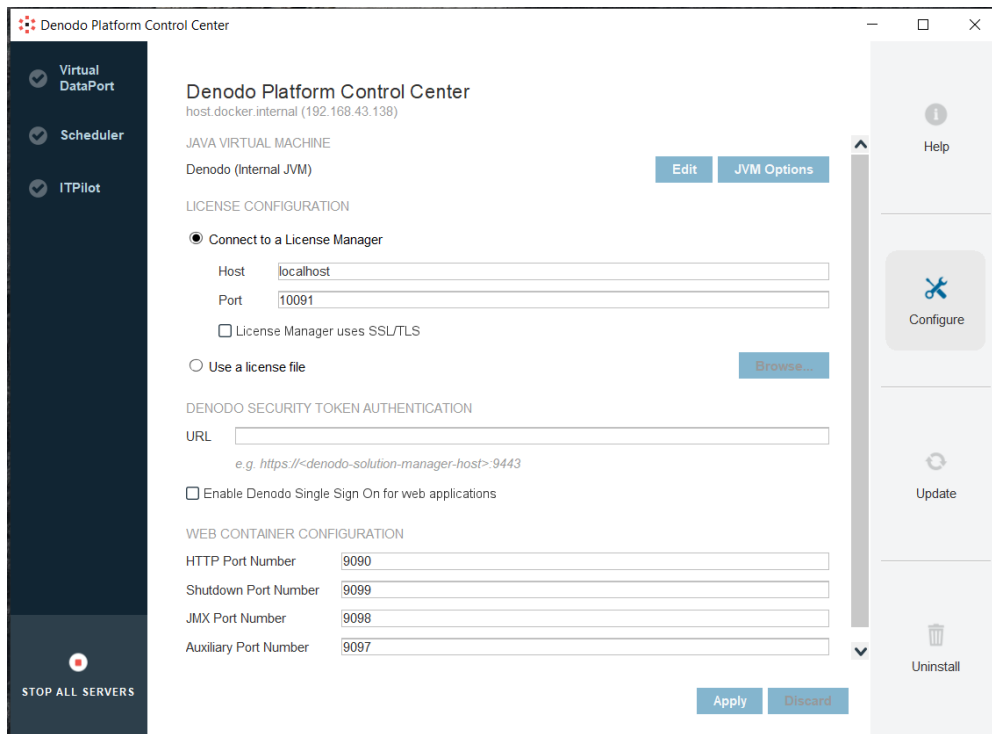
In the next step you have to select the modules to be installed. This tutorial covers every module so we suggest installing them all, but you only need Virtual DataPort to get started.



You can leave the rest of the options with their default values and complete the installation.

Once the installation is complete, you can choose to create a desktop shortcut that can be used to start the **Denodo Control Center**.

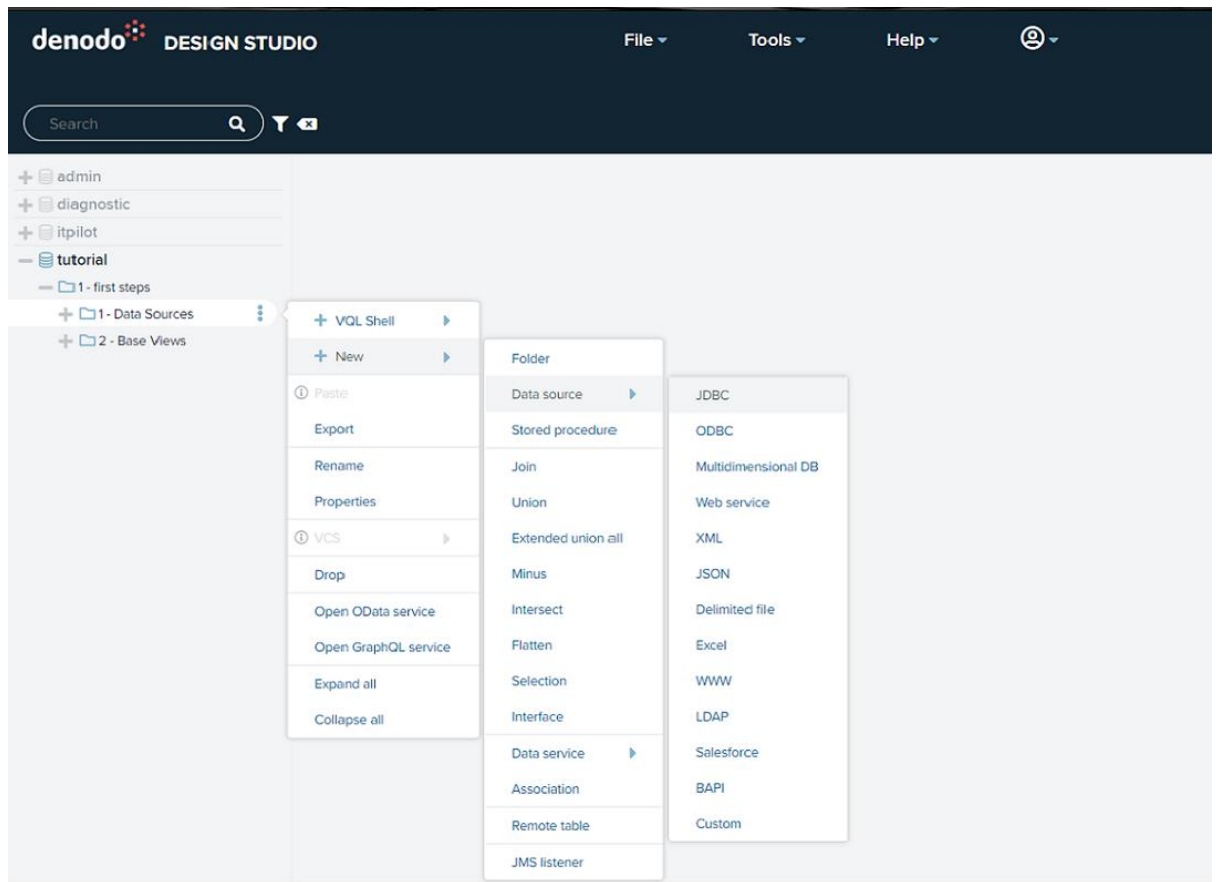
If you did not select any Denodo license during the installation process, you can do it from the **Denodo Control Center** when you first start the program. (By first clicking on the **Configure** button and then selecting **Use a license file**, a new dialog will be opened to select the license file and confirm its installation.



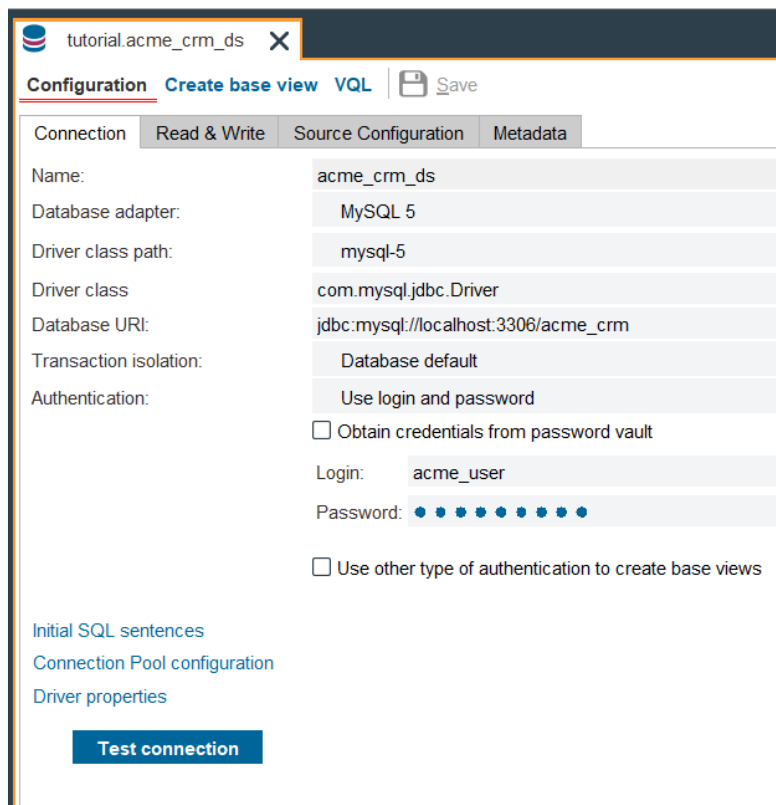
Importing relational databases:

MySQL schema has been attached

1. I had created one folder, 1. Data Sources. Right-click on the "1 - Data Sources" folder and select "New > Data source > JDBC".

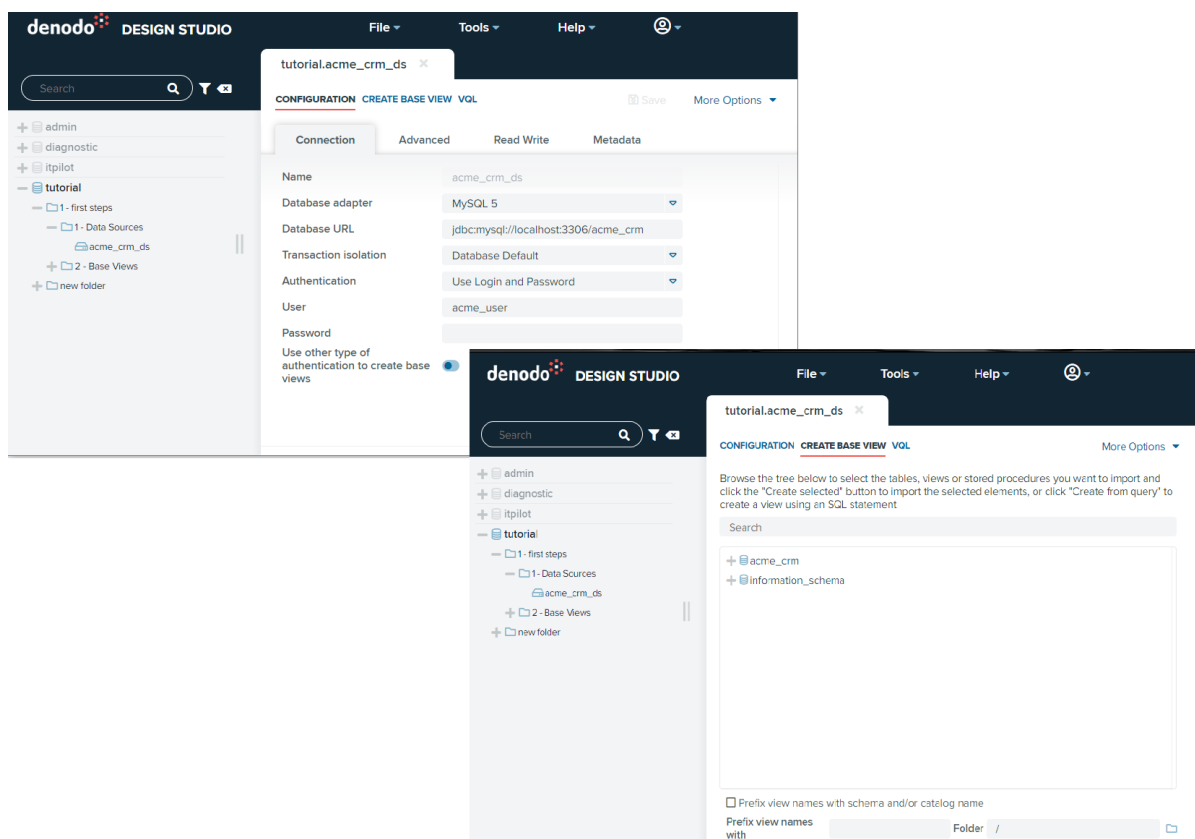


This is the window that appeared

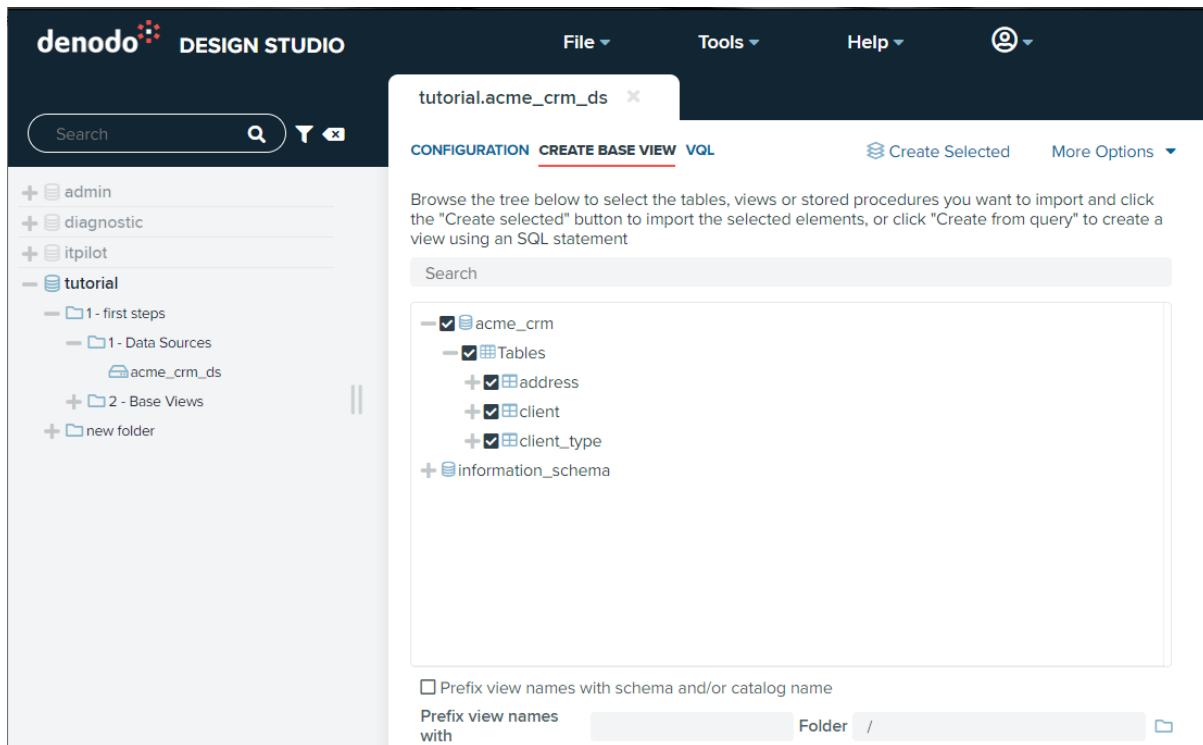


2. Then, I named the data source as **acme_crm_ds**.
3. Selected "**MySQL 5**" as Database Adapter.
4. Changed the Database URI to match your MySQL installation, for example, **jdbc:mysql://localhost:3306/acme_crm**
5. Entered the login and password fields: **acme_user / acme_user**.
6. Clicked on "**Test connections**".
7. Click on the "  Save" button.
8. Click the "Create base view" button at the top.

The Administration Tool will show the introspected schema of the relational database:

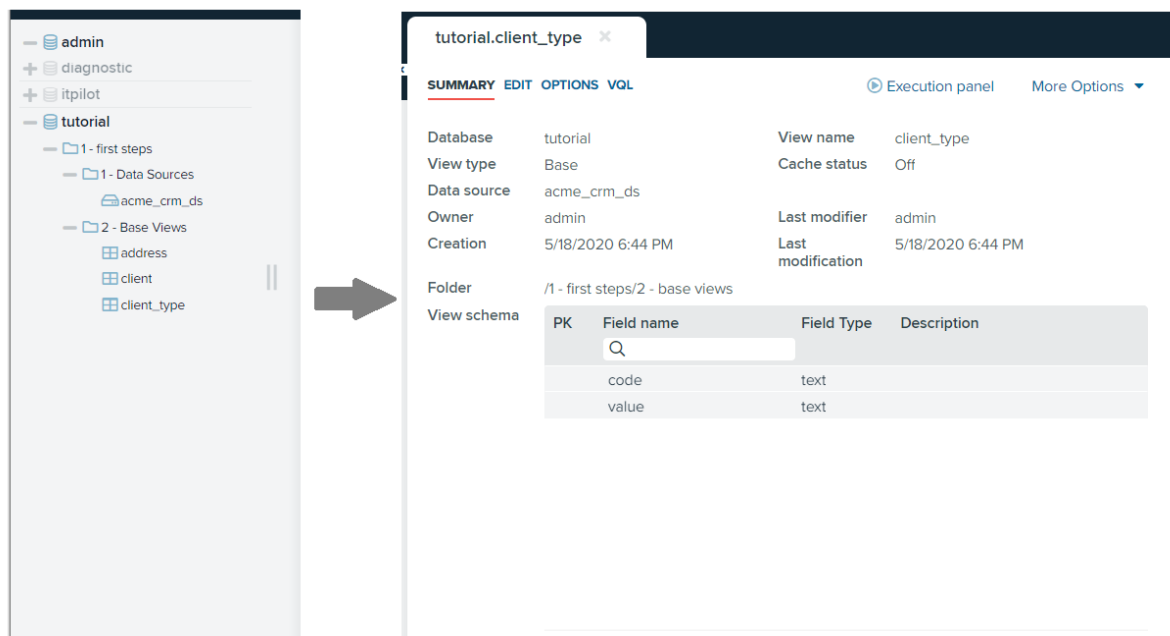


To incorporate some of the tables into the Denodo virtual schema, you have to check the box near the tables or views you want to import. In this case, check **address**, **client** and **client_type** and then click on the **Create selected** button.



Later, you will be able to query these base views or combine them with other views.

When the importing process is finished, you will see the new views in the elements tree panel. If you double-click on the view name, the schema of the base view will be shown in the workspace.

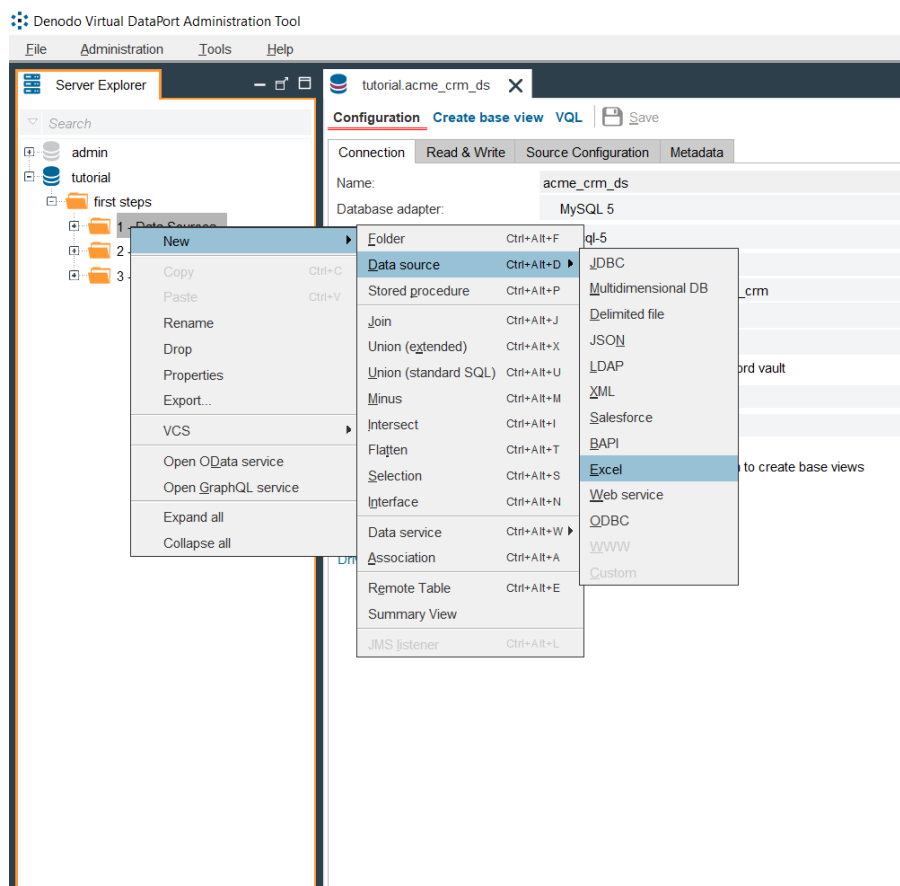


As you can see, each new base view will be listed in the Elements Tree :

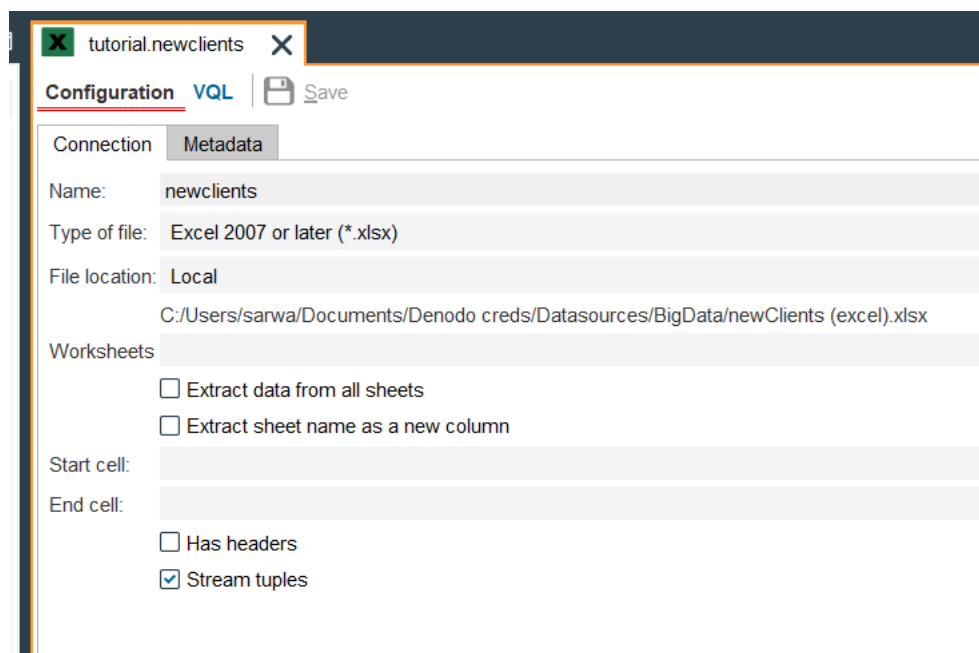
- In the folder where the view has been created ("2 - "Base Views").

Importing Data from Excel:

- . Right-click on the "1 - Data Sources" folder and select "New > Data source > Excel".



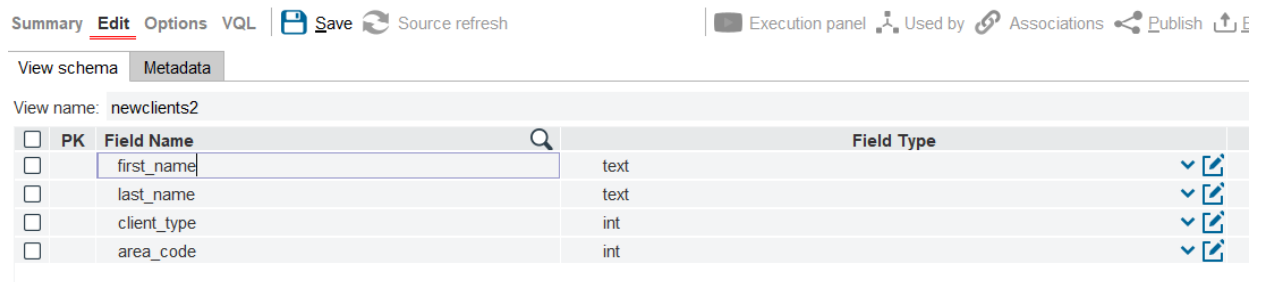
- Then, I entered the following information



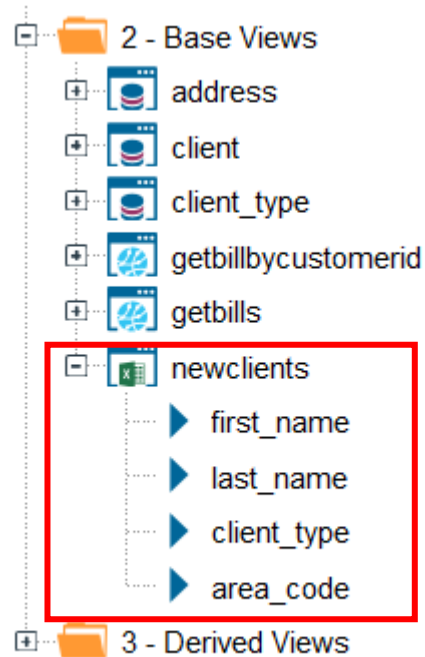
- In file location, I gave the path of the excel file in my local machine.
- I saved the changes, and clicked on *create base view*.



- Then, I assigned a name for each column as:
 - first_name
 - last_name
 - client_type
 - area_code



- In the Base View folder, the fields are entered



- Now, we have imported data from excel database

Performing SQL operations on excel table:

- After importing excel database, click on newclients under base view folder.
- Click on Execution Panel

Field Name	Field Type	Description
first_name	text	
last_name	text	
client_type	int	
area_code	int	

Owner: admin Last modifier: admin
 Creation: 20/04/2021 11:06:53 AM Last modification: 20/04/2021 11:06:53 AM
 Swap status: default Cache status: off
 Folder: /first steps/2 - base views
 Description:

- The execution panel appears with query, `SELECT * FROM newclients CONTEXT ('i18n'='au_cst', 'cache_wait_for_load'='true')`. Click on Execute.

Execute X

Quick Query Specify Where Expression

Current sentence:

```
SELECT * FROM newclients CONTEXT ('i18n'='au_cst', 'cache_wait_for_load'='true')
```

☐ Do not use cache
 ☐ Invalidate existing results
 Display rows: 150
 ☐ Open results in new tab

☐ Store results in cache
 ☐ Do not use swap
 ☐ Retrieve all rows

Conditions +

- After clicking on execute, this is the result which is obtained.

Execute X Query Results X

Results Execution Trace Stop Refresh Save Query: SELECT * FROM newclients2 LIMIT 150 CONTEXT ('i18n'='au_cst', 'cache_wait_for_load'='true') TRACE

Total rows received: 150 (shown 150)

first_name	last_name	client_type	area_code
Regina	Buckley	1	328
Mercedes	Moore	2	750
Iola	Wheeler	2	442
Martha	Norman	3	986
Jorden	Burnett	2	409
Roth	Miles	1	102
Deirdre	Marshall	3	405
Melyssa	Navarro	2	107
Nell	Fitzpatrick	1	649
Steel	Mack	3	54
Uta	Gibson	1	753
Lance	Hester	1	481

- You can also specify where condition. Click on + sign next to Conditions. Enter column name and the expression. Click on execute.

Execute X Query Results X

Quick Query Specify Where Expression

Current sentence:

```
SELECT * FROM newclients WHERE client_type = 1 CONTEXT ('i18n'='au_cst', 'cache_wait_for_load'='true')
```

☐ Do not use cache
 ☐ Invalidate existing results
 Display rows: 150
 ☐ Open results in new tab

☐ Store results in cache
 ☐ Do not use swap
 ☐ Retrieve all rows

Conditions +

client_type = 1

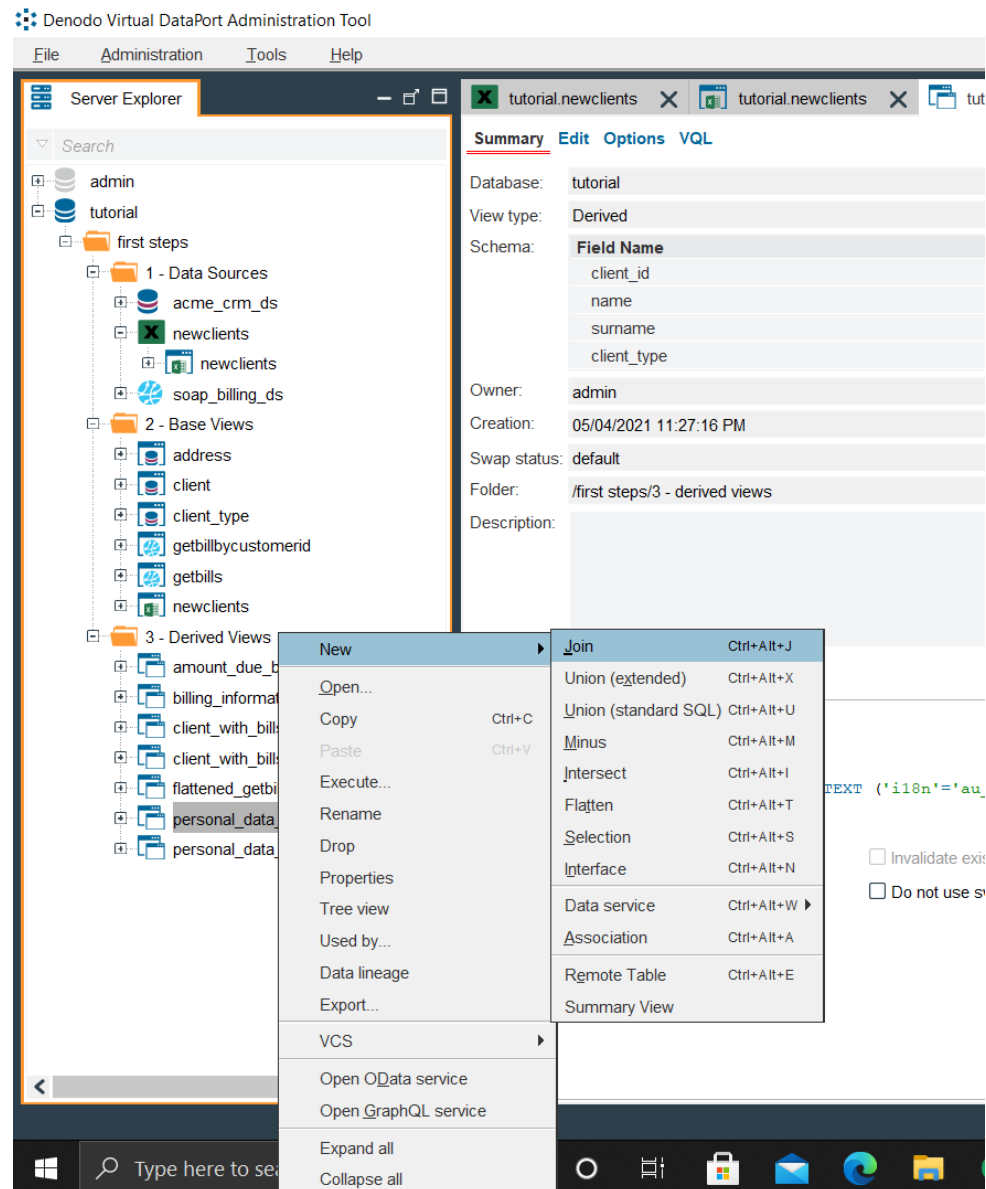
- This is the output of it:

Execute X		Query Results X	
Results	Execution Trace	Stop Refresh Save	Query: SELECT first_name AS first_name, last_name AS last_name, client_type AS client_type, area_code AS area_code
Total rows received: 150 (shown 150)			
△ ▽			
first_name	last_name	client_type	area_code
Regina	Buckley	1	328
Roth	Miles	1	102
Nell	Fitzpatrick	1	649
Uta	Gibson	1	753
Lance	Hester	1	481
Dorothy	Morton	1	328
Rudyard	Edwards	1	71
Xenos	Bowers	1	204
Ivan	Oneil	1	906
Jael	Mathews	1	985
Salvador	Blackburn	1	710
Wyoming	Ferguson	1	101

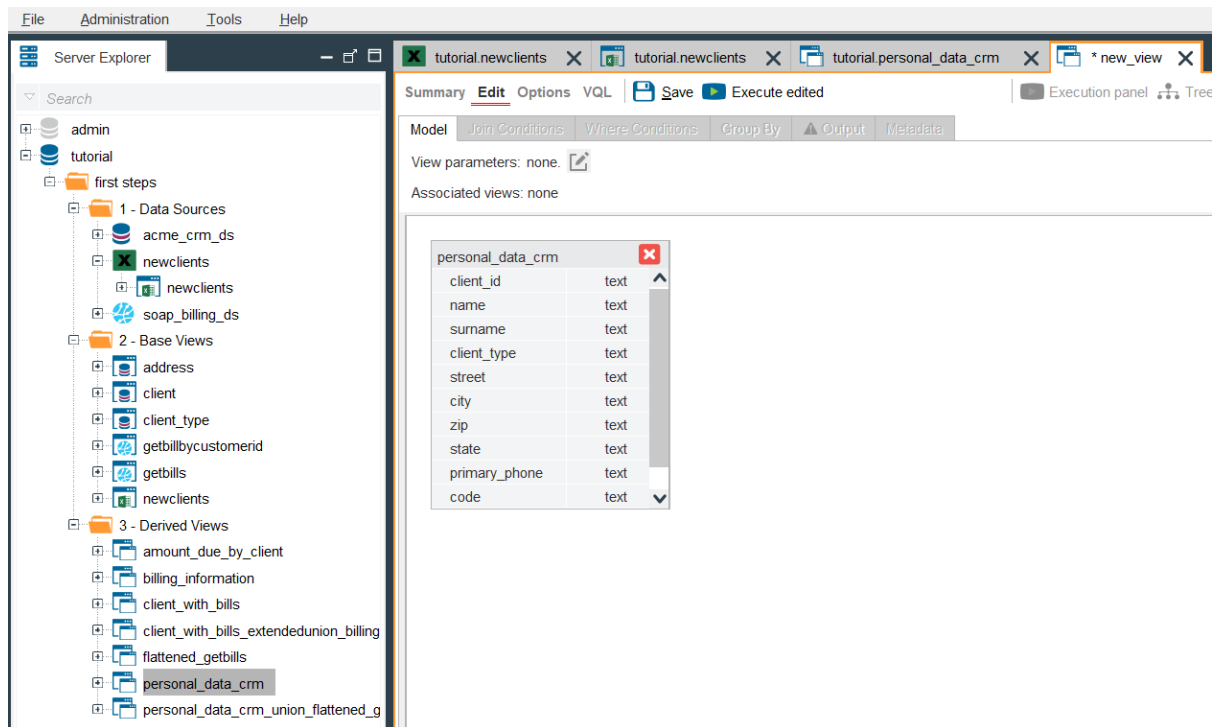
Performing Join operation between heterogenous databases

personal_data_crm is a derived view, consisting of join operation between tables of MySQL database. I had performed join operation on personal_data_crm view and newclients view (which is derived from excel database)

- Right click on personal_data_crm under derived view folder. Select "New > Join".

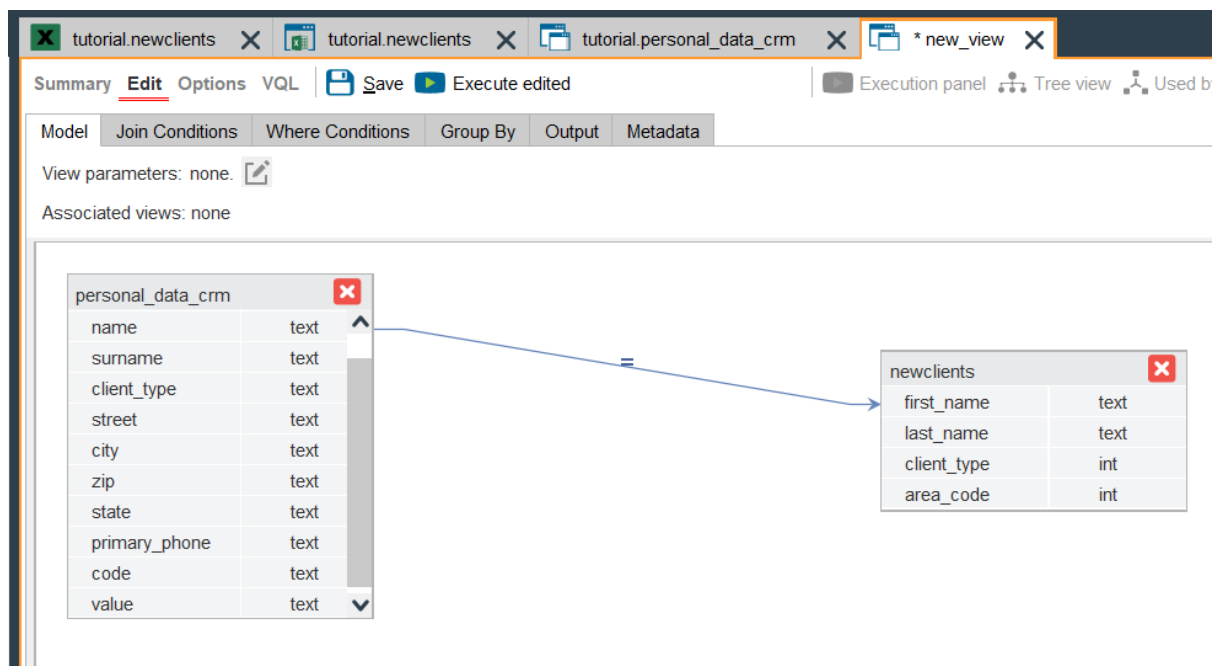


- This window appears



Drag and drop newclients from Base Views to the Join View wizard Model tab.

- I used **personal_data_crm.name = newclients.first_name** as my join condition. So drag & drop a line from the *name* field in the **personal_data_crm** view to the *first_name* field in the **newclients** view.



- Name the join view, **personal_data_crm_join_newclients** in Output tab.
- Save the view.

- Click on Execution Panel. Then click on Execute on Execution Panel.

The screenshot shows the Databricks interface. At the top, there are tabs for 'tutorial.newclients', 'tutorial.newclients', 'tutorial.personal_data_crm', and 'tutorial.personal_data_crm_join_newclients'. The 'tutorial.personal_data_crm_join_newclients' tab is active. Below the tabs, there is a 'Summary' tab and an 'Execution panel' tab, which is highlighted with a red box. The 'Execution panel' tab shows a table with columns 'Field Name', 'Field Type', and 'Description'. The table contains rows for 'client_id', 'name', 'surname', and 'client_type'. Below the table, there are fields for 'Owner', 'Creation', 'Swap status', 'Folder', and 'Description'. The 'Execute' button is highlighted with a red box.

- Result after performing join operation

The screenshot shows the Databricks interface with the 'Query Results' tab selected. The query is: `SELECT * FROM personal_data_crm_join_newclients LIMIT 150 CONTEXT ('i18n'='au_cst', 'cache_wait_for_load'='true') TRACE`. The results are displayed in a table with 13 columns: client_id, name, surname, client_type, street, city, zip, state, primary_phon... code, value, first_name, last_name, and area_code. The table contains 150 rows of data, showing a join operation between two datasets.

Performing Union operation between heterogenous databases

getbills a view, derived from a web service. So, I performed a union operation on personal_data_crm view and getbill view in the similar way as Join operation.

- Model Tab:

Summary Edit Options VQL Save Execute edited Execution panel Tree view Used by Data li

Model Associations Where Conditions Group By Output Metadata

View parameters: none.

personal_data_crm	
client_id	text
name	text
surname	text
client_type	text
street	text
city	text
zip	text
state	text
primary_phone	text
code	text

flattened_getbills	
ssn	text
billing_end_date	text
phone_center	text
due_date	text
balance	text
billing_period_id	text
billing_start_date	text
billing_id	text
return_customer_id	text
package_id	text

- Associations Tab:

Model Associations Where Conditions Group By Output Metadata

Associations list:

 personal_data_crm.client_id = flattened_getbills.return_customer_id

- Execution Panel:

Execute Query Results

Quick Query Specify Where Expression Execute Query plan

Current sentence:

```
SELECT * FROM personal_data_crm_union_flattened_getbills CONTEXT ('i18n='au_cst', 'cache_wait_for_load'='true')
```

☐ Do not use cache
 ☐ Invalidate existing results
 Display rows: 150
 ☐ Open results in new tab

☐ Store results in cache
 ☐ Do not use swap
 ☐ Retrieve all rows

Conditions +

- Output:

Execute

Query Results

ResultsExecution Trace

Stop

Refresh

Save

Query: SELECT * FROM personal_data_crm_union_flattened_getbills LIMIT 150 CONTEXT ('i18n'=au_cst, 'cache_wait_for_load'=true) TRACE

Total rows received: 150 (shown 150)

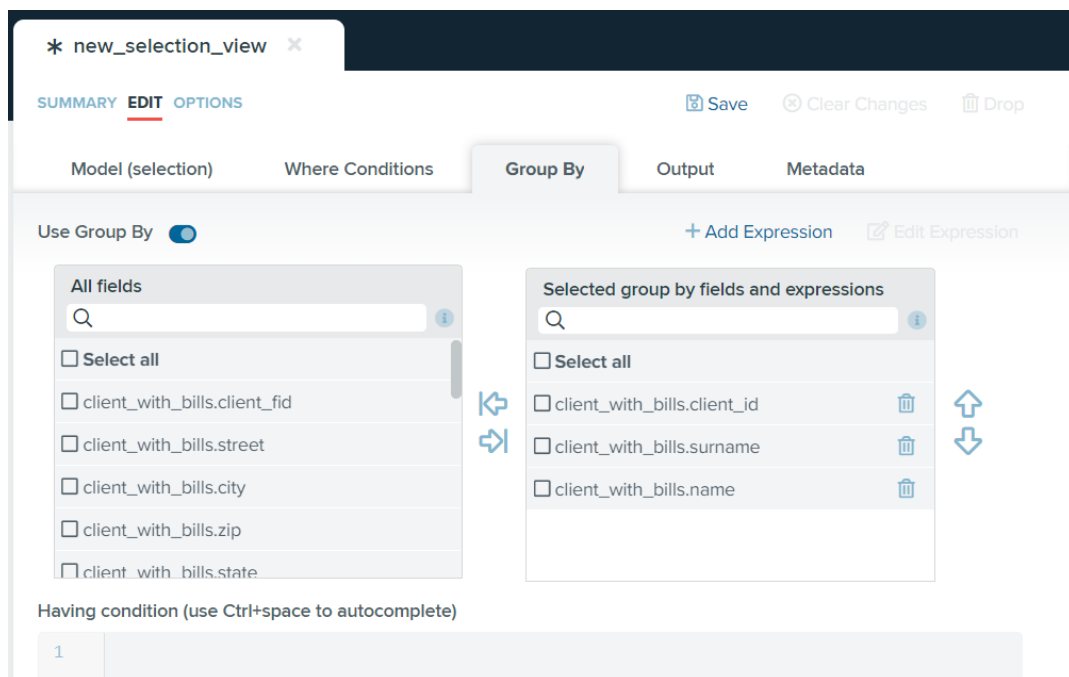
client_id	name	surname	client_type	street	city	zip	state	primary_p...	code	value	ssn	billing_end...	phone_cen...	due_date	balanc
C001	John	Smith	01	3989 Middlef...	San Jose	94085	CA	(408) 813-9...	01	Residential	<null>	<null>	<null>	<null>	<null>
C001	John	Smith	01	3989 Middlef...	San Jose	94085	CA	(408) 813-9...	02	Business	<null>	<null>	<null>	<null>	<null>
C002	Pat	Smith	01	2189 Capitol...	San Jose	94085	CA	(408) 473-9...	01	Residential	<null>	<null>	<null>	<null>	<null>
C002	Pat	Smith	01	2189 Capitol...	San Jose	94085	CA	(408) 473-9...	02	Business	<null>	<null>	<null>	<null>	<null>
C003	Jack	Smith	02	754 Southa...	San Jose	94085	CA	(408) 322-2...	01	Residential	<null>	<null>	<null>	<null>	<null>
C003	Jack	Smith	02	754 Southa...	San Jose	94085	CA	(408) 322-2...	02	Business	<null>	<null>	<null>	<null>	<null>
C004	Frank	Smith	02	1900 Univer...	San Jose	94085	CA	(408) 322-8...	01	Residential	<null>	<null>	<null>	<null>	<null>
C004	Frank	Smith	02	1900 Univer...	San Jose	94085	CA	(408) 322-8...	02	Business	<null>	<null>	<null>	<null>	<null>
C005	Maria	Smith	01	3149 Morris ...	San Jose	94085	CA	(408) 320-9...	01	Residential	<null>	<null>	<null>	<null>	<null>
C005	Maria	Smith	01	3149 Morris ...	San Jose	94085	CA	(408) 320-9...	02	Business	<null>	<null>	<null>	<null>	<null>
C006	Mary	Smith	01	777 San Ant...	San Mateo	94401	CA	(408) 320-8...	01	Residential	<null>	<null>	<null>	<null>	<null>
C006	Mary	Smith	01	777 San Ant...	San Mateo	94401	CA	(408) 320-8...	02	Business	<null>	<null>	<null>	<null>	<null>
C007	W.	Smith	01	866 Colorad...	San Mateo	94401	CA	(408) 857-9...	01	Residential	<null>	<null>	<null>	<null>	<null>
C007	W.	Smith	01	866 Colorad...	San Mateo	94401	CA	(408) 857-9...	02	Business	<null>	<null>	<null>	<null>	<null>
C008	S	Smith	02	3603 Lupine ..	San Mateo	94401	CA	(408) 856-8...	01	Residential	<null>	<null>	<null>	<null>	<null>
C008	S	Smith	02	3603 Lupine ..	San Mateo	94401	CA	(408) 856-8...	02	Business	<null>	<null>	<null>	<null>	<null>
C009	Wendy	Smith	02	1656 Channi...	San Mateo	94401	CA	(408) 322-2...	01	Residential	<null>	<null>	<null>	<null>	<null>
C009	Wendy	Smith	02	1656 Channi...	San Mateo	94401	CA	(408) 322-2...	02	Business	<null>	<null>	<null>	<null>	<null>
C010	Yi-Ning	Smith	01	879 Newell Pl	San Mateo	94401	CA	(510) 322-1...	01	Residential	<null>	<null>	<null>	<null>	<null>
C010	Yi-Ning	Smith	01	879 Newell Pl	San Mateo	94401	CA	(510) 322-1...	02	Business	<null>	<null>	<null>	<null>	<null>
C011	Zi H	Smith	01	797 Allen Ct	Palo Alto	94303	CA	(650) 320-8	01	Residential	<null>	<null>	<null>	<null>	<null>

Performing Aggregation operation between heterogenous databases

getbills a view, derived from a web service. So, I performed a Aggregation operation on personal_data_crm view and getbill view in the similar way as Join operation.

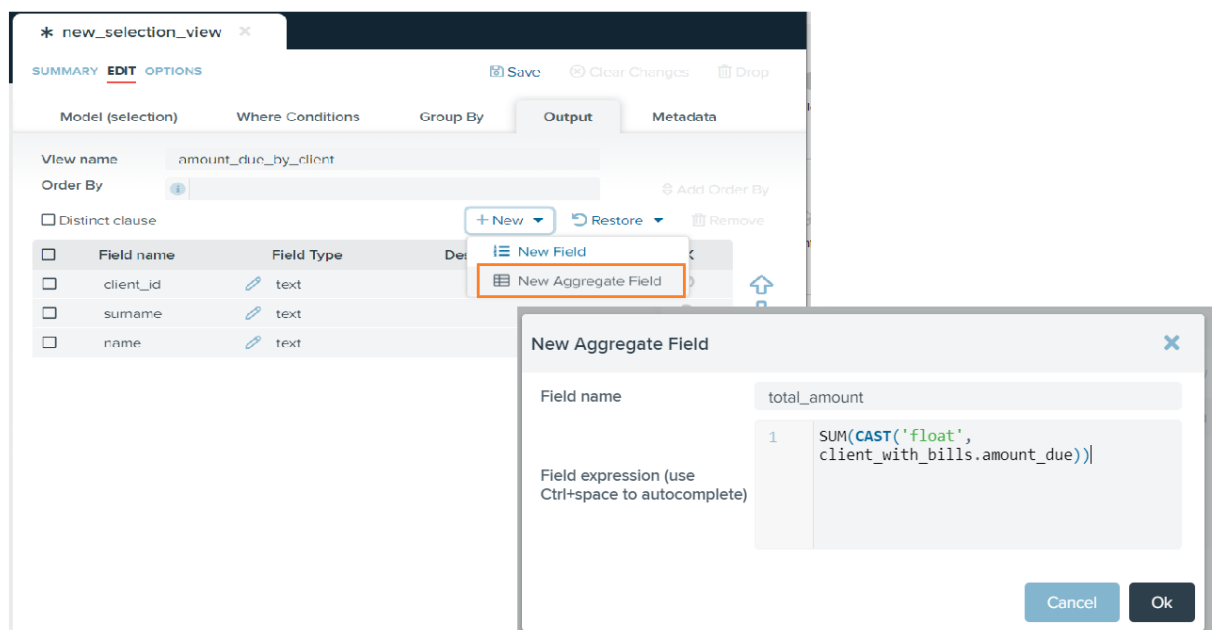
To create the new aggregation view I followed these steps:

1. Click on the **client_with_bills** view and select **New > Selection**.
2. In the **Projection / Selection View** wizard go to the **Group by** tab and enable the **Use group by** Checkbox.
3. Click on **Ok** in the pop-up dialog.
4. From the list of fields on the left **Add** the following to the **Selected group by fields and expressions** box: *name*, *surname* and *client_id*.



5. Go to the **Output** tab and rename the view to *amount_due_by_client*.
6. Now, you have to create a new field that computes the addition of the amount due for each client. Click on **New > New aggr. field** button and define a new field called **total_amount** that will be calculated using the **SUM** aggregation function and the *amount_due* field.

SUM(CAST('float', client_with_bills.amount_due))



7. Click **Ok** to add the new field to the view.
8. And click  to confirm the creation of the view.

Now, if you execute the new view, you will get as result one record per customer with the total billing balance:

tutorial.amount_due_by_client

SUMMARY EDIT OPTIONS VQL

Execution panel Used By Data Lineage More Options

Database tutorial View name amount_due_by_client

Execute Query Results

RESULTS EXECUTION TRACE

Refresh Save Copy Trace to Clipboard

80 rows 4 columns received `SELECT * FROM tutorial.amount_due_by_client CONTEXT('cache_wait_for_load' = 'tr`

Chariene	Liao	C017	<null>
Benjamin	Liangxu	C016	<null>
Yifan	Liang	C015	<null>
Sara	Liang	C014	<null>
Dongming	Liang	C013	<null>
Der S	Liang	C012	31.99
Zi H	Smith	C011	<null>
Yi-Ning	Smith	C010	<null>
Wendy	Smith	C009	<null>
S	Smith	C008	<null>
W.	Smith	C007	<null>
Mary	Smith	C006	<null>
Maria	Smith	C005	75.99
Frank	Smith	C004	<null>
Jack	Smith	C003	204.98
Pat	Smith	C002	21.99
John	Smith	C001	3.99